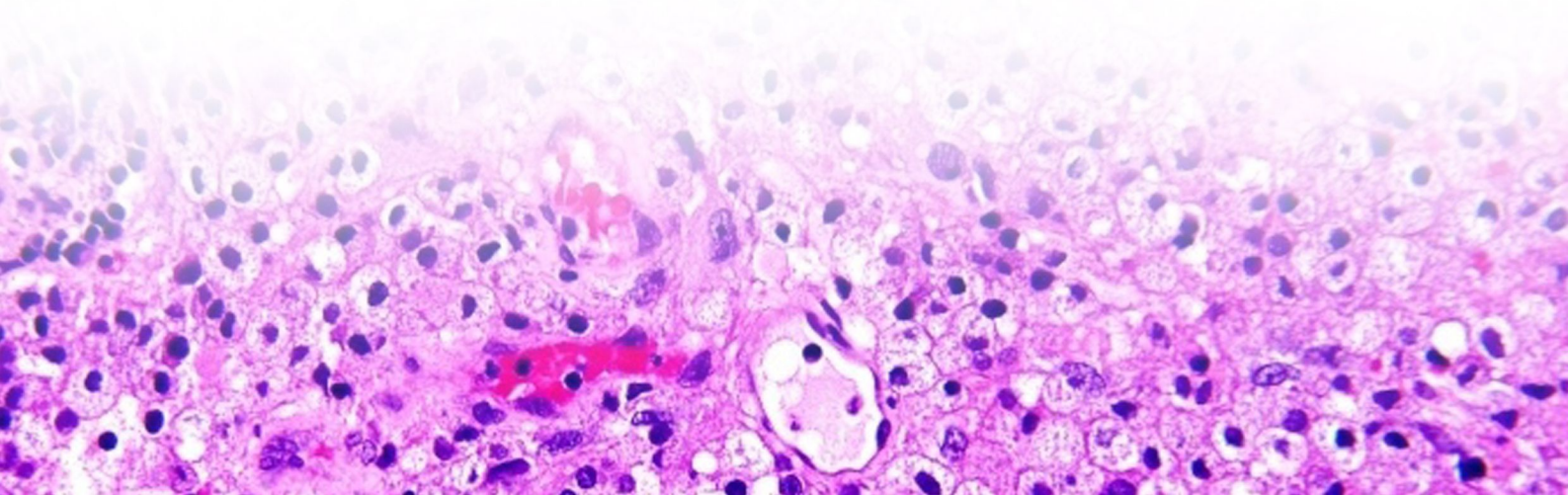
Student Notes

PATHOLOGY

1st Edition

Unit 9

Environmental Diseases



Preface

Agam is a group of budding medicos, who are currently doing their under graduation in various Medical Colleges across Tamil Nadu and Pondicherry. The group was initiated on 18th November 2017, in the vision of uniting medicos for various social and professional causes.

We feel delighted to present you Agam Pathology notes prepared by Agam Divide and Rule 2020 Team to guide our fellow medicos to prepare for university examinations.

This is a reference work of 2017 batch medical students from various colleges. The team took effort to refer many books and make them into simple notes. We are not the authors of the following work. The images used in the documents are not copyrighted by us and is obtained from various sources.

Dear readers, we request you to use this material as a reference note, or revision note, or recall notes. Please do not learn the topics for the 1st time from this material, as this contain just the required points, for revision.

Acknowledgement

On behalf of the team, Agam would like to thank all the doctors who taught us Pathology. Agam would like to whole heartedly appreciate and thank everyone who contributed towards the making of this material. A special thanks to Vignesh M, who took the responsibility of leading the team. The following are the name list of the team who worked together, to bring out the material in good form.

- Fredrick Snowin William
- Swathikrishna S
- Pavithra Devi K
- Ishwarya J
- Aakash R
- Vignesh. M
- Varsha L
- Muthamil Selvi E
- Kaushik N R

ENVIRONMENTAL AND NUTRITIONAL DISEASES

SHORT NOTES:

1. Vitamin A deficiency
2. Vitamin D deficiency
3. Wilson Disease
4. Obesity
5. Lead Poisoning
6. Silicosis

SHORT ANSWERS:

1. Conditions associated with PEM
2. Lead poisoning
3. Vitamin C deficiency
4. Examples of trace elements and their deficiency

UPDATES

SHORT NOTES

1. VITAMIN A DEFICIENCY

The normal range of vitamin A is 28–86 µg/dL.

FUNCTIONS OF VITAMIN A

- Maintenance of normal vision
- Cell growth and differentiation
- Lipid metabolism

CAUSES OF VITAMIN A DEFICIENCY

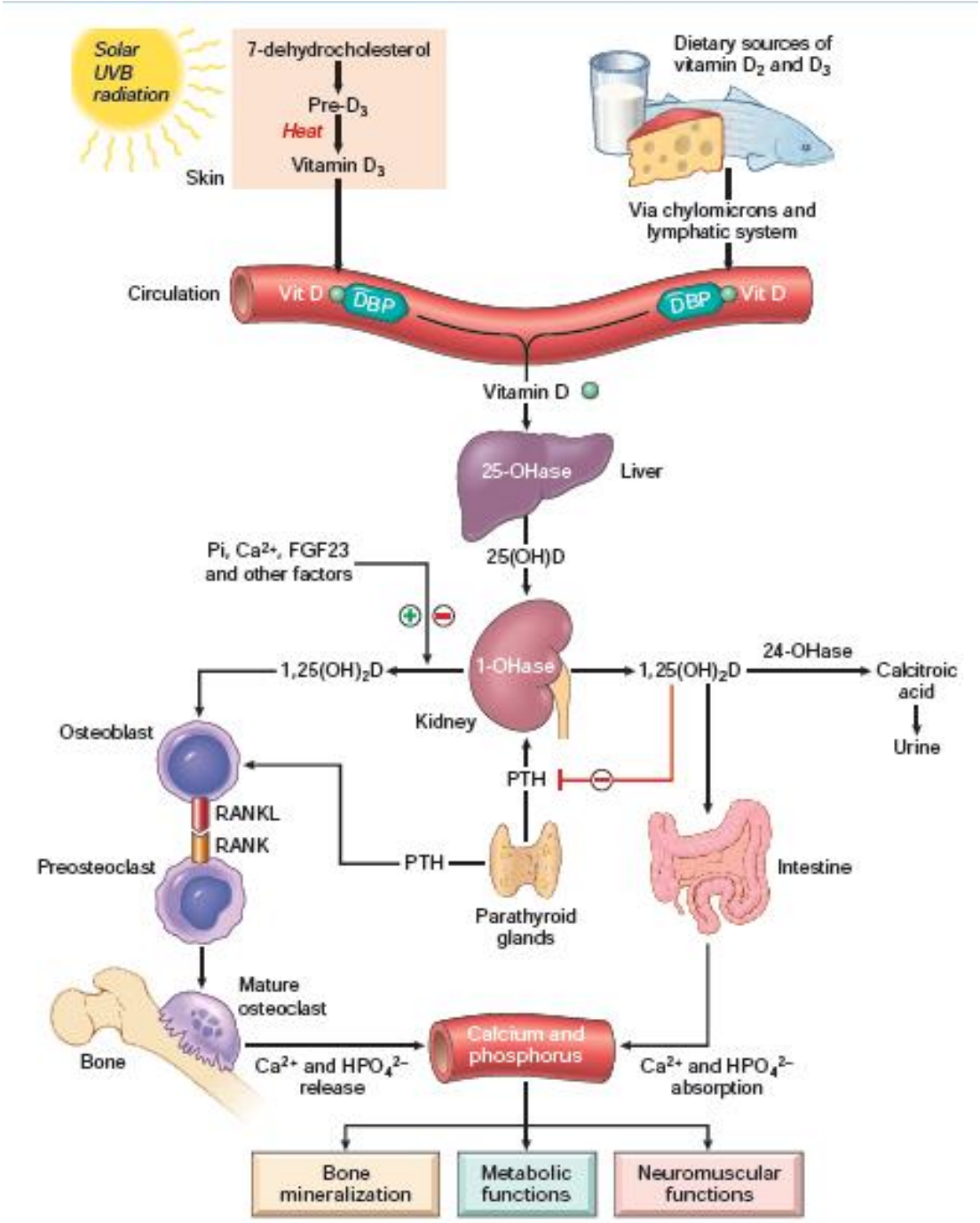
- Undernutrition
- Malabsorption of fat
- Malabsorption syndrome (Celiac disease, Crohn disease, Colitis)
- Bariatric surgery
- Laxative overuse

DEFICIENCY DISORDERS

- **Night blindness** - component of rhodopsin
- **Epithelial metaplasia and keratinization**
- **Xerophthalmia - Xerosis conjuctivae**
- **Bitot spots** (keratin debris cornea)
- **Keratomalacia** - destruction of cornea
- **Squamous metaplasia**
  - Respiratory tract (respiratory infections)
  - Urinary tract (renal stones)
- **Follicular or papulary dermatosis** - hyperkeratinization of epidermis
- **Immune deficiency**

2. VITAMIN D DEFICIENCY:

VITAMIN D METABOLISM:



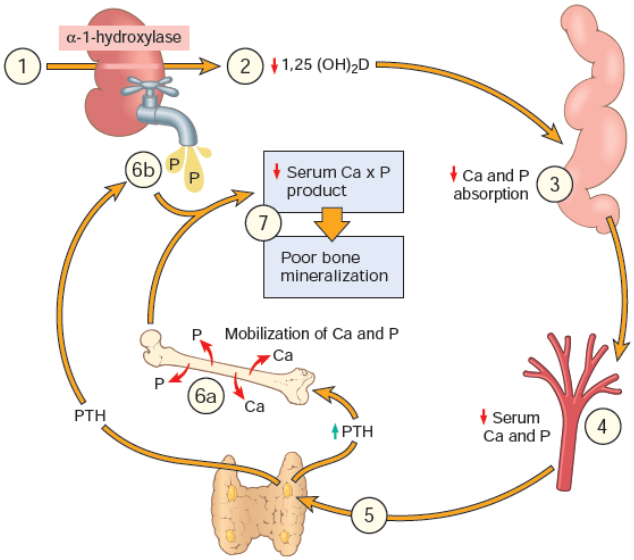
- Vitamin D SUPPLEMENTS increase lymphocyte count, circulating cytokines, clearance of bacteria from sputum in tuberculosis patients
- Macrophages, keratinocytes can produce mild amount of vitamin D BY USING CYP27B and vitamin D receptors' are also present in them

VITAMIN D:

- **Functions of vitamin D**:
  - Plasma calcium and phosphorous level
  - Mineralization of bone
  - Neuromuscular transmission
- **Normal range** : **20ng/ml - 50ng/ml**
- **Deficiency: <20 ng/ml**
- **Causes of Vitamin D deficiency:**
  - Dietary deficiency
  - Limited exposure to sunlight
  - Frequent pregnancy
  - Renal disorder
  - Inherited disorder
  - Malabsorption disorders
- **Aftermath of Vitamin D deficiency**:
  - **Rickets** – children
  - **Osteomalacia** - adults
  - Mild disease like vitamin D insufficiency (hip fracture)
  - 30-40% increase in colon, prostate and breast cancer

VITAMIN D DEFICIENCY:

CAUSES:



1. ↓ Substrate for the renal 1α-hydroxylase
2. Yielding a deficiency of 1,25 (OH)2D and
3. Deficient absorption of Ca and P from gut with
4. Consequently depressed serum levels of both.
5. The hypocalcemia activates the parathyroid
6. Causing mobilization of Ca and P from bone

6a. PTH induces wasting of phosphate in the urine

6b. And calcium retention.

7. As a result, the serum levels of calcium are normal or nearly normal, but phosphate levels are low; hence, mineralization is impaired

RICKETS

The following sequence ensues in rickets:

- **Overgrowth of epiphyseal cartilage** due to:
  - ✓ Inadequate provisional calcification
  - ✓ Failure of the cartilage cells to mature and disintegrate
- **Persistence of distorted, irregular masses of cartilage**, which project into the marrow cavity
- **Deposition of osteoid matrix** on inadequately mineralized cartilaginous remnants
- **Disruption of the orderly replacement of cartilage** by osteoid matrix, with enlargement and lateral expansion of the osteochondral junction
- **Abnormal overgrowth of capillaries and fibroblasts** in the disorganized zone resulting from microfractures and stresses on the inadequately mineralized, weak, poorly formed bone
- **Deformation of the skeleton** due to the loss of structural rigidity of the developing bones

FEATURES:

- **Non - Ambulatory stage of infancy**:
  - The head and chest sustain the greatest stresses.
  - **Craniotabes:** The softened occipital bones may become flattened, and the parietal bones can be buckled inward by pressure; with the release of the pressure, elastic recoil snaps the bones back into their original positions
  - **Rachitic Rosary:** An excess of osteoid produces:
    - ✓ Frontal bossing
    - ✓ Squared appearance to the head.
  - **Pigeon Breast Deformity:**
    - ✓ Due to overgrowth of cartilage or osteoid tissue at the costochondral junction
    - ✓ The weakened metaphyseal areas of the ribs are subject to the pull of the respiratory muscles
    - ✓ Thus they bend inward, creating anterior protrusion of the sternum.
- **Ambulatory stage of infancy**
  - Deformities affect the spine, pelvis, and tibia, and causes **lumbar lordosis** and **bowing of the legs**

OSTEOMALACIA

- The lack of vitamin D deranges the normal bone remodelling that occurs throughout life.
- The newly formed osteoid matrix laid down by osteoblasts is inadequately mineralized, thus producing the **excess of persistent osteoid** that is characteristic of osteomalacia.
- The contours of the bone are not affected
- The bone is weak and vulnerable to gross fractures or micro fractures, which are most likely to affect vertebral bodies and femoral necks.
- The unmineralized osteoid appears as a thickened layer of matrix (which stains pink in hematoxylin and eosin preparations) arranged about the more basophilic, normally mineralized trabeculae.

3. WILSON'S DISEASE

- Autosomal recessive inherited disease.
- Accumulation of copper in tissues- mutation of the ATP7B gene

TRIAD OF FEATURES:

- Liver cirrhosis
- Bilateral degeneration of the basal ganglia
- Kayser-Fleischer rings → Greenish-brown pigmented rings in the periphery of the cornea.

BIOCHEMICAL ABNORMALITIES:

- Decreased serum ceruloplasmin
- Increased hepatic copper in liver biopsy.
- Increased urinary excretion of copper.

MORPHOLOGIC FEATURES:

- Fatty change.
- Acute and chronic active hepatitis.
- Submassive liver necrosis and macronodular cirrhosis.
- Mallory bodies are present in some cases.

4. OBESITY

- BMI>30
- Accumulation of adipose tissue that impairs health
- Obesity depends upon
    - Genetic
    - Nutritional
    - Environmental and
    - Psychological signals
- The neuro-humoral mechanism controls appetite and satiety by 3 components
    - Afferent system
    - Arcuate nucleus
    - Efferent system

LEPTIN

- Produced by ob genes in fat cells (adipocyte)
- Act on ob receptors (type 1 cytokines receptors)-jak-stat pathway
- Activation of pomc/cart neuron and inhibition of npy/agrp neuron
- Cause increase energy expenditure
- Obesity is caused by loss of function of leptin system, mutation of mcr-4, haploinsufficiency of BDNF
- When leptin level increases it causes increased physical activity, thermogenesis, increase proinflammatory cytokines, hematopoesis, ymphopoesis

ADIPONECTIN

- Fatty acid oxidation in muscles
- Synthesized in adipocytes and bind to Adipo R1 and Adipo R2 receptors in liver, skeletal muscles
- Called as fat burning molecule and guardian angel against obesity
- It is a complex of 3-6 monomers

GUT HORMONES: They include

- **Ghrelin** (synthesized by stomach and arcuate nucleus)
- **Peptide YY** (ileum and colon)
- **Pancreatic polypeptide, insulin and amylin** (pancreas)

Ghrelin will increase food intake by stimulating NPY / AgRP NEURONS

PYY reduce energy intake (PYY level decreases causing hyperphagia) act by stimulating POMC/CART NEURON

AMYLIN will reduce food intake by stimulating POMC/CART NEURONE

ADIPOCYTES

- They produce TNF ,IL-6,IL-1,IL-18,chemokines and steroid hormones which create a pro inflammatory state by increasing C reactive protein
- Their number will be high in obese individuals
- Loss of fat mass causes their shrinkage

GUT MICROBIOME

- Alterations in gut microbiome affect obesity
- They control breakdown of diet absorption of nutrients ,epithelial integrity and inflammation

CONSEQUENCES OF OBESITY

- Type 2 diabetes
- Cardiovascular diseases
- Metabolic syndrome-
  - Visceral or intraabdominal adiposity
  - Insulin resistance
  - Hyperinsulinemia
  - Glucose tolerance
  - Hypertension
  - Hypertriglyceremia
  - Low HDL cholesterol
- Non-alcoholic fatty liver disease
- Cholelithiasis
- Hypoventilation
- Hyper somnolence
- Osteoarthritis
- Cancer

5. LEAD POISONING

CAUSES:

- **Children**
  - Chewing of lead containing objects
  - Lead paint flakes from walls
- **Adult**
  - **Occupational exposure-**spray painting, mining & extraction of lead.
  - **Accidental exposure**
    - ✓ Contaminated water supply
    - ✓ House freshly coated with lead paint
    - ✓ Sniffing of lead

CLINICAL FEATURES:

- Absorbed through the gastrointestinal tract:
- Bones (90%), teeth, nails and hair-increased bone densities ('lead lines')
- Brain, liver, kidneys and bone marrow (10%)

Nervous system:

- Children-lead encephalopathy, oedema of brain, flattening of gyri and compression of ventricles.
- Adult- demyelinating peripheral motor neuropathy → wristdrop and footdrop

Haematopoietic system:

- Microcytic hypochromic anaemia → inhibition of delta-aminolevulinic acid dehydrogenase and ferroketolase
- Basophilic stippling of erythrocytes.

Kidneys:

- Lead nephropathy → accumulation of intranuclear inclusion bodies consisting of lead-protein complex in the proximal tubular cells.

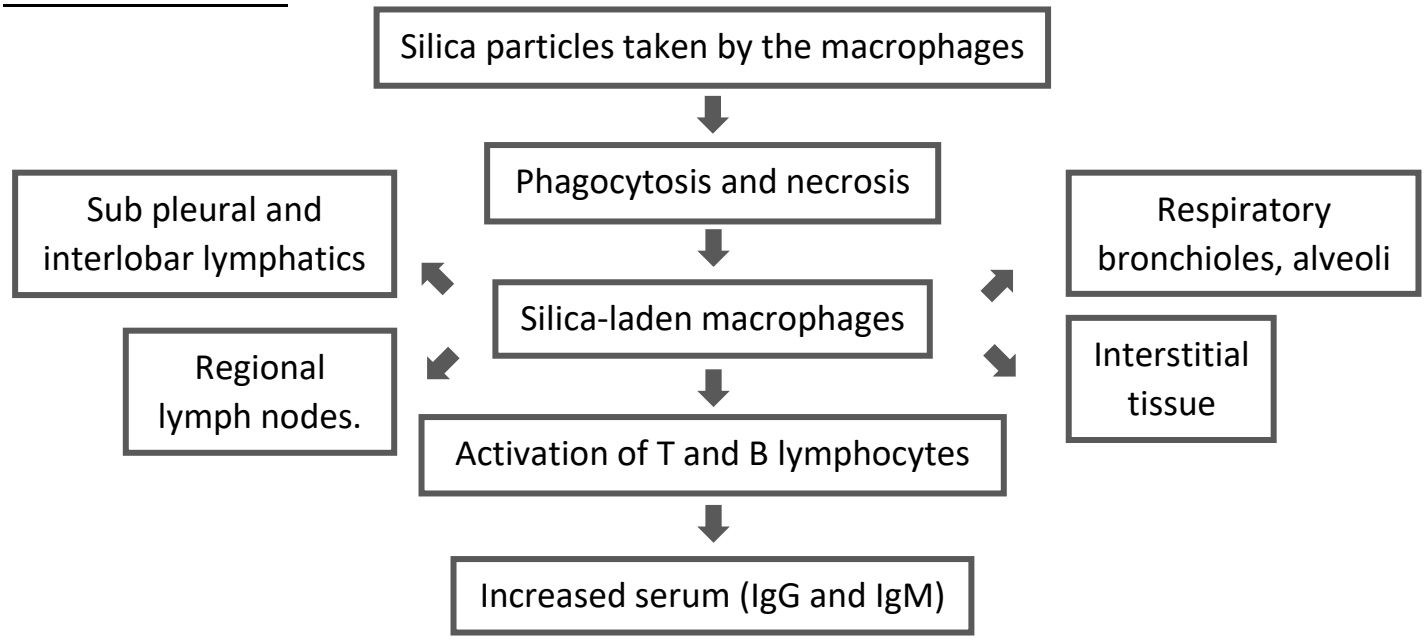
Gastrointestinal tract:

- Acute abdomen → lead colic.

6. SILICOSIS

Also known as 'knife grinders' lung

PATHOGENESIS:



MORPHOLOGIC FEATURES:

- Well-circumscribed, hard, fibrotic nodules, 1- 5 mm in diameter, scattered throughout the lung parenchyma.
- Simultaneous deposition of coal dust and may develop calcification.
- Pleura → grossly thickened& adherent to the chest wall.
- Lesions may undergo ischaemic necrosis → cavitation, or complicated by TB & rheumatoid pneumoconiosis.

HISTOLOGICAL FEATURES:

- Silicotic nodules → central hyalinised material,scanty cellularity& dust.
- Hyalinised centre → surrounded by concentric laminations of collagen& enclosed by cellular connective tissue, dust-filled macrophages, few lymphocytes and plasma cells.
- Calcium deposits may be seen.
- Collagenous nodules → cleft-like spaces- numerous birefringent particles of silica.
- Intervening lung parenchyma → hyperinflation or emphysema.

CLINICAL FEATURES:

- Dyspnoea
- Complications such as pulmonary tuberculosis, rheumatoid arthritis (Caplan's syndrome) and cor pulmonale may occur.

SHORT ANSWERS:

1. PEM (PROTEIN ENERGY MALNUTRITION)

- PEM is the common cause of childhood deaths in poor countries
- 2 main primary PEM syndromes are marasmus and kwashiorkor
- Secondary PEM occurs in chronically ill and in patients with advanced cancer
- Malnutrition is determined according to the BMI (normal range 18.5 to 25 kg/m2)
- Kwashiorkor is characterized by hypoalbuminemia, generalized edema, fatty liver, skin changes, and defects in immunity. It is caused by diets low in protein but normal in calories.
- Marasmus is characterized by emaciation resulting from loss of muscle mass and fat with relative preservation of serum albumin. It is caused by diets severely lacking in calories-both protein and nonprotiens.

2. VITAMIN C DEFICIENCY

- Vitamin C deficiency leads to scurvy
- Major consequences of vitamin C deficiency is caused by impaired formation of collagen:
- Poor vessel support results in bleeding tendencies
- Inadequate synthesis of osteoid
- Impaired wound healing

3. EXAMPLES OF TRACE ELEMENTS AND THEIR DEFICIENCY

- **Iron** – hypo chromic microcytic anaemia
- **Iodine** – goitre and hypothyroidism
- **Copper** –muscle weakness, neurologic defects and abnormal collagen cross linking
- **Fluoride** –dental caries
- **Selenium** –myopathy and keshan's disease
- **Zinc** –acrodermatitis enteropathica, anorexia, diarrhoea, impaired night vision, depressed wound healing and infertility

UPDATES:

1. RADON

- It is a radioactive gas derived from uranium and is widely present in soil and in homes
- It can cause lung cancer in uranium miners.
- It is also suspected that low-level chronic exposures in the home increase lung cancer risk, particularly in those who smoke tobacco.
- Radon is the number one cause of lung cancer among nonsmokers
- Overall, radon is the second leading cause of lung cancer.
- It is suspected that lower levels of household radon may also contribute to lung cancer development, particularly in those who also smoke tobacco.

2. BURNING OF ORGANIC MATERIALS

- Smoke from burning of organic materials, containing various oxides of nitrogen and carbon particulates
- It is an irritant that predisposes exposed persons to lung infections and may contain carcinogenic polycyclic hydrocarbons.

3. FLINT WATER CRISIS

- A dramatic case of lead contamination of drinking water occurred in Flint, Michigan, in 2014–2016.
- The Flint water crisis occurred when the source of water supply to the city was changed from Lake Huron to the Flint River.
- Because water from the Flint River had a higher chloride concentration than the lake waters, it leached lead from century-old lead pipes.
- This caused an increase in lead levels in tap water above the acceptable limit of 15 parts per billion (ppb) in about 25% of the homes and in some cases as high as 13,200 ppb.
- As a result, 6000 to 12,000 residents developed very high lead levels in their blood.

4. EFFECT OF NICOTINE ON FETUS:

- Recent studies indicate that in addition to being addictive, nicotine has other untoward effects.
- These include effects on the fetus, as nicotine, exposure affects fetal brain development and contributes to preterm birth and still birth.

5. NEUROTRANSMISSION BY COCAINE:

- Cocaine facilitates neurotransmission in the CNS, where it blocks the reuptake of dopamine, and at adrenergic nerve endings, where it blocks the reuptake of both epinephrine and norepinephrine while stimulating the presynaptic release of norepinephrine.
- The euphoria is due to its enhancement of brain dopamine activity, especially in the so-called mesolimbic dopamine reward pathway.

6. NON-MALIGNANT EFFECTS OF SMOKING

- Agents in smoke have a direct irritant effect on the tracheobronchial mucosa, producing inflammation and increased mucus production (bronchitis).
- Cigarette smoke also causes the recruitment of leukocytes to the lung, with increased local elastase production and subsequent injury to lung tissue, leading to emphysema, chronic bronchitis, and chronic obstructive pulmonary disease

7. ELECTRONIC CIGARETTES:

- Electronic cigarettes (e-cigarettes) simulate cigarette smoking by delivering vaporized nicotine and flavourings.
- The use of flavoured e-cigarettes, called **vaping**, has been on the increase in recent years, especially among young adults.
- While for several years after the introduction of e-cigarettes no significant untoward effects were recorded.
- But starting in the summer of 2019 an outbreak of vaping-associated acute lung injury occurred in the United States.
- With the advent of e-cigarettes and vaping, an increasing number of users are inhaling aerosolized THC (tetrahydrocannabinol), a practice that has been associated with severe lung injury in some users, possibly due to the presence of adulterants.
- Paradoxically, some heavy users develop cannabis hyperemesis syndrome, marked by intractable nausea and vomiting that only remits with cessation of use.

8. EFFECTS OF MARIJUANA

- Marijuana use causes euphoria and a sense of relaxation.
- It causes a heightened sensory perception (e.g., brighter colours), laughter, altered perception of time, and increased appetite.

9. MARIJUANA USE DISORDER

- Chronic use of marijuana, especially if the exposure started during adolescence, gives rise to marijuana use disorder.
- Dependence in such individuals is manifested by withdrawal symptoms such as mood and sleep difficulties that may last up to 2 weeks when not taking the drug.
- A small subset of those who develop dependence may become addicted— individuals who cannot stop using the drug even though it interferes with many aspects of their life.

10. LSD AND ECTASY:

- Lysergic acid diethylamide (LSD) is the most potent hallucinogen known.
- LSD has unpredictable effects on mood, affect, and thought, sometimes leading to bizarre and dangerous behaviours.
- Ecstasy (3,4-methylenedioxymethamphetamine [MDMA]); and **bath salts**, synthetic cathinones that are chemically related to khat, a widely used stimulant in East Africa.
- Chronic use of Ecstasy may deplete the CNS of serotonin, potentially leading to sleep disorders, depression, anxiety, and aggressive behaviour.

11. ROLE OF HYPOTHALAMUS

- The hypothalamus is the master regulator of energy homeostasis.
- It receives inputs from the periphery about the state of energy stores
- If the energy stores are inadequate, it triggers anabolic circuits, and if they are adequate, catabolic circuits are activated.
- The effect of the **anabolic circuits** is to increase intake of food and reduce energy expenditure
- The effect of **catabolic circuits** reduce food intake and increase energy expenditure.

12. LIPOSTAT

- The two sides of the energy equation, intake and expenditure, are finely regulated by neural and hormonal mechanisms
- Apparently, this fine balance is controlled by an internal set-point, or “lipostat,”
- Lipostat senses the quantity of energy stores (adipose tissue) and appropriately regulates food intake as well as energy expenditure.

13. SEVERE ACUTE MALNUTRITION (SAM)

- Protein Energy Malnutrition is now renamed as Severe Acute Malnutrition
- The WHO defines Severe Acute Malnutrition (SAM) as a state characterized by a weight for height ratio that is 3 standard deviations below the normal range.

14. ROLE OF GUT MICROBIOME IN SAM:

- There is a substantial difference in the microbial flora of children with SAM when compared with the gut microbiome of normal children.
- The microbiome are not merely the consequences of SAM but play a role in their causation.
- Malnutrition was induced in host mice by fecal transplants from children with SAM but not well-nourished children.

15. SECOND ORDER NEURON IN ENERGY BALANCE

- Neurons bearing **melanocortin receptors 3 and 4** (MC3/4R) → receive signals from first-order POMC/CART neurons → catabolic pathway
- Neurons bearing **Y1 and Y5 receptors** → receive signals from first-order NPY/AgRP neurons → anabolic pathway

16. ADIPONECTIN AND CANCER

- Adiponectin secretion from adipose tissue is reduced in obese individuals.
- Adiponectin suppresses cell proliferation and promotes apoptosis.
- It does so in part by promoting the actions of p53 and p21.
- In obese individuals these antineoplastic actions of adiponectin may be compromised.

17. INFLAMMATION AND OBESITY

- Markers of inflammation, such as CRP and proinflammatory cytokines like TNF, are often elevated in obese persons, in particular people with central obesity.
- The basis for the inflammation is uncertain; both a direct proinflammatory effect of excess circulating lipids and increased release of cytokines from fat-laden adipocytes have been proposed.
- Whatever the cause, it is thought that chronic inflammation may contribute to many of the complications of obesity including insulin resistance, metabolic abnormalities, thrombosis, cardiovascular disease, and cancer.

18. CANCER AND OBESITY

There is an increased incidence of certain cancers in overweight people, including:

- Cancers of the esophagus, thyroid, colon, and kidney in men
- Cancers of the esophagus, endometrium, gallbladder, and kidney in women.

19. INSULIN AND LEPTIN

- Insulin, like leptin, exerts anorexigenic responses.
- Both POMC/CART and NPY/AgRP neurons express insulin receptors.
- However, while insulin can mimic the actions of leptin, most of the evidence suggests the primacy of leptin in the regulation of energy homeostasis.

20. EFFECT OF DIET ON DISEASE

- Restricting sodium intake reduces hypertension.
- Dietary fiber, or roughage, resulting in increased fecal bulk, provides a preventive effect against diverticulosis of the colon.
- Caloric restriction has been demonstrated to increase lifespan in experimental animals.

21. VEGETABLE OILS AND FISH OILS

- Vegetable oils (e.g., corn and safflower oils) and fish oils contain polyunsaturated fatty acids and are good sources of such cholesterol-lowering lipids.
- Fish oil fatty acids belonging to the omega-3, or n-3, family have more double bonds than the omega-6, or n-6, fatty acids found in vegetable oils.
- Dietary supplements of omega-3 fatty acids or consumption of oily fish had little or no effect on cardiovascular disease (ischemic heart disease, stroke).

22. ONE LINERS

- Cardiovascular effects of Arsenic include hypertension and prolonged Q-Tc interval with ventricular arrhythmias.
- Opioid drugs of abuse include prescription drugs such as oxycodone, hydrocodone, fentanyl, tramadol, and methadone.
- There is no increase in the risk for ovarian cancer by using estrogen-progestin combination
- In women, low to moderate intake of alcohol (12 oz beer or 5 oz of wine) incurs a slightly higher risk of breast cancer.
- Older adults (above 65 years) are much more likely to suffer from adverse drug reactions

- Apart from other cancers, cigarette smoking is also associated with Carcinomas of liver and colon
- Smoking increases the risk of type 2 diabetes, rheumatoid arthritis, age-related macular degeneration, ectopic pregnancy, and erectile dysfunction.
- **Alcohol Abuse Disorder (AUD)** is a chronic relapsing brain disease characterized by an impaired ability to stop or control alcohol use despite adverse social, occupational, or health consequences.
- The hyperthermia resulting from 'Malignant hyperthermia' has a mortality rate of approximately 80% if untreated, but this falls to less than 5% if the condition is recognized and muscle relaxants are administered promptly.
- **Frostbite** - Direct effects of hypothermia are probably mediated by physical disruptions within cells by high salt concentrations caused by the crystallization of intracellular and extracellular water
- With a loss of fat, measured skinfold thickness (which includes skin and subcutaneous tissue) is reduced
- Vitamin A supplementation can reduce morbidity and mortality rates from diarrhea by 15% and 28%, respectively.
- Obesity is defined as an accumulation of adipose tissue that is of sufficient magnitude to impair health.
- Obstructive sleep apnea and consequent right-sided heart failure is strongly associated with obesity.
- Evidence suggests that excess fatty acids can cross the blood–brain barrier and enter the hypothalamus, where they are sensed by microglial cells.

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